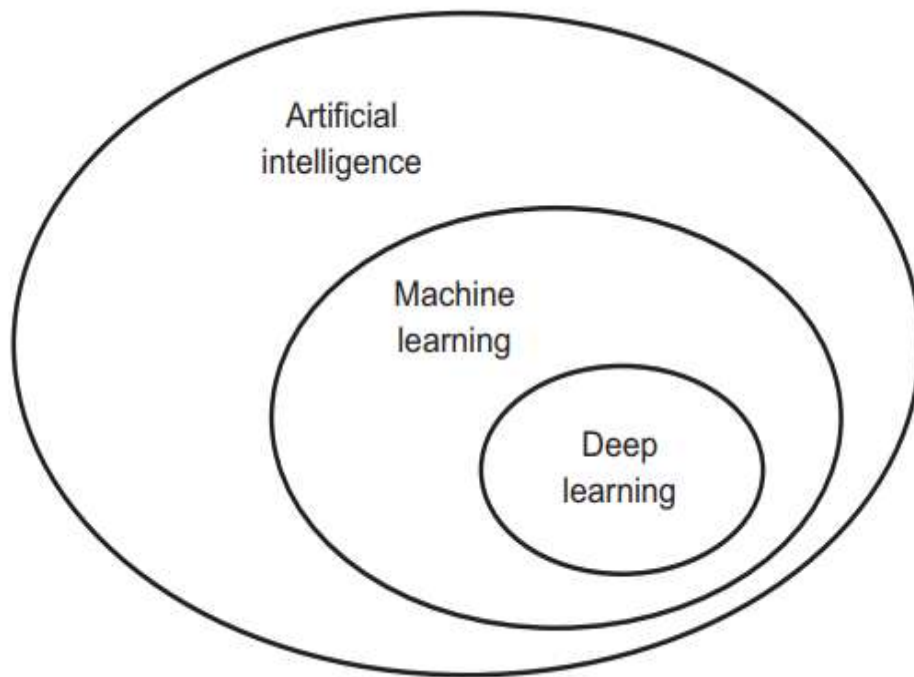


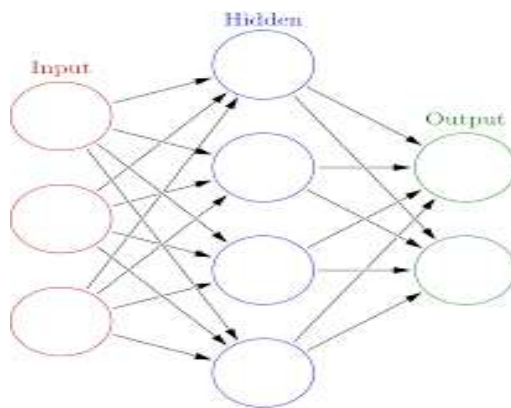
Introduction to Neural Network

Artificial intelligence, machine learning, and deep learning



Neural Networks

- Machine learning models that mimic the complex functions of the human brain.
- These models consist of interconnected nodes or neurons.



Evolution of Neural Networks

- 1940s-1950s: **First mathematical model for** artificial neurons
- 1960s-1970s: [Perceptrons](#) are simple single-layer networks.
- 1980s: The development of backpropagation.
- the 1990s: “AI winter”. Popular in different industries
- **2000s**: Availability of larger datasets.
- 2010s-Present: The landscape of machine learning has been dominated by deep learning models.

Understanding Neural Networks in Deep Learning

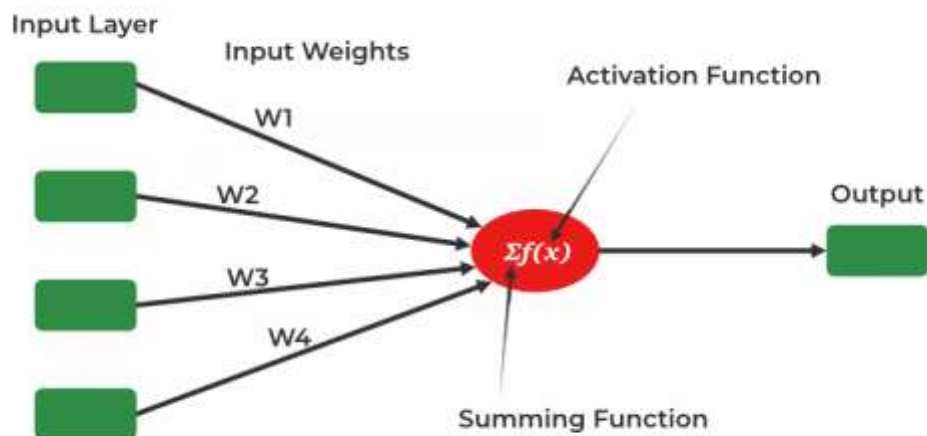
These networks are built from several key components:

1. **Neurons**: The basic units that receive inputs.
2. **Connections**: Links between neurons that carry information.
3. **Weights and Biases**: These parameters determine the strength and influence of connections.
4. **Propagation Functions**: Mechanisms that help process and transfer data across layers of neurons.
5. **Learning Rule**: The method that adjusts weights and biases over time to improve accuracy.

Learning in neural networks follows a structured, three-stage process:

1. **Input Computation**: Data is fed into the network.
2. **Output Generation**: The network generates an output based on the current parameters.
3. **Iterative Refinement**: The network refines its output by adjusting weights and biases, gradually improving its performance on diverse tasks.

Simple single-layer networks



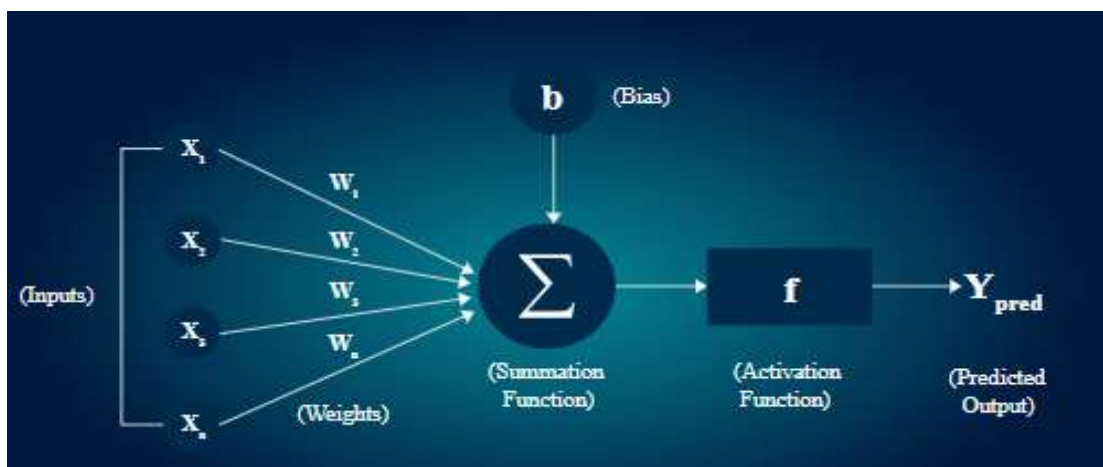
Working of Neural Networks

Forward Propagation

1. Input
2. Weight+bias
3. Activation function - **introduces non-linearity into the system**

Backpropagation

1. Loss Calculation
2. Gradient Updation- optimizer
3. Weight updation



Iterative Refinement

Iteration

This process of forward propagation, loss calculation, backpropagation, and weight update are repeated for many iterations over the dataset.

Over time, this iterative process reduces the loss, and the network's predictions become more accurate.

Limitations of single layer

- Only applicable on linearly separable data
- Can't capture complex pattern because of single layer structure.